YANG HU

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EDUCATION

New York University — Sep. 2024 - May 2026
Master of Science, Computer Science

University of California, Santa Barbara — Sep. 2020 - Mar. 2024
Bachelor of Science, Mathematics/Statistics and Data Science — Major GPA: 3.75, CS GPA: 3.94

SKILLS

Languages	Fluent with C++, Python, SQL, R; Experienced with HTML/CSS, HUGO
Frameworks	PyTorch, OpenCV, Scikit-Learn, Langchain, Faiss, Spark, PostgreSQL, Django, Nginx
Tools	Google Cloud Platform, AWS, Travis CI, Redis, Git, SVN, Shell, LaTeX, Agile

EXPERIENCE

R&D Software Engineering Intern — May 2024 - Aug. 2024
Unity — *Shanghai, China*

- Scaled the RAG database for Unity's AI assistant by 10× through a workflow that systematically gathers inputs from various Unity forums and enhances data quality with an LLM pipeline.
- Built MuseBench, an evaluation platform that combines a fine-tuned LLM-as-a-judge with a benchmark dataset, cutting AI agent assessment time by 95% and drastically reducing human review efforts.
- Developed a pipeline with a locally deployed LLM to automatically parse and extract critical error messages from Unity Cloud Build log files typically exceeding 100k lines.

Machine Learning Undergraduate Researcher — Jun. 2023 - Mar. 2024
The WAVES Lab, University of California, Santa Barbara — *Santa Barbara, CA*

- Deployed state-of-the-art deep learning models for image segmentation on large-scale satellite imagery.
- Proposed a novel computationally efficient model that addresses key segmentation challenges in remote sensing.
- Recognized with the Schmidt Family Foundation Research Mentorship Award.

PROJECTS

VocationalNYC — Connecting New Yorkers to Local Training Opportunities — Jan. 2025 - Ongoing

- Engineered a robust backend with Django to power user authentication and program listings, leveraging PostgreSQL for scalable relational data management and WebSockets for seamless bi-directional communication.
- Integrated the Google Maps API to deliver an interactive search experience with proximity-based sorting.
- Deployed the full-stack app on AWS Elastic Beanstalk with Nginx as a reverse proxy, and Travis CI for CI/CD.

Information Retrieval System and Retrieval-Augmented Generation — Sep. 2024 - Dec. 2024

- Built a ground-up search engine system on the MSMACRO dataset using an inverted index optimized with block compression and a Document-at-a-Time query processor employing BM25 ranking.
- Revamped the retrieval pipeline by reconstructing the dataset using HNSW, boosting the F1 Score by 25%.
- Investigated the effectiveness of multiple reordering strategies in mitigating the "Lost in the Middle" biases in RAG pipelines through large-scale key-value retrieval experiments.

Semantic Segmentation by Pixel-level Time Series Classification — Jan. 2023 - Mar. 2024

- Implemented various pixel-level time series classification models utilizing satellite data from Google Earth Engine.
- Evaluated the transferability and adaptability of the trained model across various locations and timeframes.

Few-shot Instance Segmentation for Remote Sensing — Jun. 2023 - Dec. 2023

- Developed a novel strategy named STC leveraging the Segment Anything Model and Vision Transformer for instance image segmentation in remote sensing, reducing manual labeling and training costs by 70%.

PUBLICATIONS

[1] Z. Wu, J. Li, **Yang Hu**, *et al.* "Compacter: A Lightweight Transformer for Image Restoration". *ACM Multimedia*, 2024.

[2] **Yang Hu**, K. Caylor, A. Boser. "Segment-then-Classify: Few-shot instance segmentation for environmental remote sensing". *NeurIPS Workshop on Tackling Climate Change with Machine Learning*, 2023.